1 Train-only single-head SPA-Net

This candidate-four assignment study uses Delphes simulation and nested train-source-group OOF only. No validation, test, observed data, or Run-2 expected-sensitivity claim is made. Exactly $3b \to \geq 4b$ is the primary QCD projection; direct $\geq 4b$ QCD is secondary closure only.

The learned-minus-geometric exact-pairing difference has median 0.01840; its 68% interval is [0.01311, 0.02320]. All resampleable values and comparisons use the same complete-source-member draw registry. Classification differences are reported without improvement claims whenever their paired interval includes zero.

Model	Weighted AUC	Nominal Z_A	Syst. Z_A
Inclusive BDT	$0.7710^{+0.0102}_{-0.0091}$	$0.06370^{+0.00658}_{-0.00557}$	$(3.776^{+0.938}_{-0.751}) \times 10^{-5}$
Single-head SPA-Net	$0.6569^{+0.0163}_{-0.0183}$	$0.04390^{+0.00351}_{-0.00292}$	$(1.692^{+0.296}_{-0.244}) \times 10^{-5}$

Model	S/B	Background N_{eff}
Inclusive BDT	$(8.495^{+1.723}_{-1.462}) \times 10^{-5}$	$28.68^{+5.46}_{-4.25}$
Single-head SPA-Net	$(4.016^{+0.638}_{-0.515}) \times 10^{-5}$	$58.39^{+7.56}_{-7.91}$

Pairing	Exact accuracy	Matchable efficiency	Ambiguity	Mass RMS [GeV]
Frozen geometric	$0.8297^{+0.0040}_{-0.0041}$	$0.6182^{+0.0033}_{-0.0039}$	$(4.276^{+2.153}_{-2.129}) \times 10^{-4}$	$22.75^{+0.45}_{-0.43}$
Single-head SPA-Net	$0.8478^{+0.0047}_{-0.0043}$	$0.6182^{+0.0033}_{-0.0039}$	$(4.276^{+2.153}_{-2.129}) \times 10^{-4}$	$24.64^{+0.49}_{-0.48}$

Metric	Nominal Δ	Median Δ	68% interval	$P(\Delta > 0)$
weighted auc	-0.1144	-0.11457	[-0.12676, -0.10235]	0.000
nominal asimov ZA	-0.019788	-0.019642	[-0.025608, -0.014257]	0.000
systematic aware asimov ZA	-2.0446e-05	-2.0498e-05	[-2.943e-05, -1.3966e-05]	0.000
signal over background	-4.4095e-05	-4.4102e-05	[-6.0664e-05, -3.0875e-05]	0.000
background effective events	29.743	29.117	[22.55, 35.697]	0.999
Pairing accuracy	0.018027	0.018395	[0.013109, 0.023201]	1.000

Outer fold	Threshold	Inner accuracy	Outer accuracy	Assignment train rows
0	0.62022	0.8462	0.8464	4421
1	0.64674	0.8400	0.8476	4732
2	0.62683	0.8299	0.8542	4404
3	0.64413	0.8452	0.8524	4766
4	0.64147	0.8466	0.8366	4753